

Fully Interpretable SECA Rule-Driven Manned-Unmanned Teaming Confrontation Strategy: A Deep Human-Machine Collaboration Framework

Chenbo Ding , Le Ru , Maolong Lv , and Li Mo

Abstract—Despite the extensive research conducted in autonomous aerial confrontation, most of these studies have focused on homogeneous Unmanned Aerial Vehicles (UAVs). Heterogeneous aerial confrontation, represented by Manned-Unmanned Teaming (MUM-T), still faces two major challenges: (i) The lack of exploration into the underlying mechanisms of human-machine collaborative combat, which makes it difficult to fully leverage the respective advantages of manned aerial vehicles and UAVs; (ii) The absence of fully interpretable safety strategies limits the practical applicability of existing research findings. To address these challenges, this paper proposes a fully interpretable MUM-T framework for beyond-visual-range aerial confrontation, driven by State-Event-Condition-Action (SECA) rules. This framework introduces, for the first time in the context of MUM-T aerial confrontation, a comprehensive mechanism for target allocation and guidance task allocation at the multi-fighter level. Additionally, at the single-fighter level, this framework designs a weapon launch strategy based on value-distance assessment and differentiated decision-making rules. Based on the SECA methodology, this paper proposes a verification approach for rule correctness and validates the proposed rules. In simulated engagements against three typical tactics, the proposed framework demonstrates favorable performance in combat effectiveness, resource management, and airspace control. Typical case analysis further confirms the full interpretability. Furthermore, the framework's overall effectiveness

is systematically validated through parameter advantage analysis, complexity and scalability analysis, and robustness analysis.

Index Terms—Manned-Unmanned Teaming, Beyond-Visual-Range aerial confrontation, autonomous decision-making, fully interpretability, human-machine deep cooperation.

I. INTRODUCTION

WITH flexibility, versatility, and low-cost advantages, Unmanned Aerial Vehicles (UAVs) have been widely employed in modern military operations such as data collection [1], surveillance [2], and communication [3]. The application of unmanned combat forces is triggering a new transformation in modern Beyond-Visual-Range (BVR) aerial confrontation [4], [5]. However, the current level of autonomous intelligence in UAVs is insufficient to handle the complex and dynamic battlefield environment [6]. The future trend lies in Manned-Unmanned Teaming (MUM-T), which can fully leverage the command and control capabilities of Manned Aerial Vehicles (MAVs) and the operational flexibility of UAVs [7], [8]. Aerial confrontation constitutes a highly complex dynamic system, particularly in multi-fighter BVR scenarios. The central challenge in MUM-T BVR aerial confrontation is to achieve optimal formation effectiveness by harnessing the distinct advantages of MAVs and UAVs, fostering deep human-machine cooperation, and designing fully interpretable strategies.

A. Related Work

In early research on aerial confrontation, rule-based approaches are the primary method. These methods guide decision-making by predefining sets of rules, offering strong interpretability and feasibility. Several studies draw on classical decision-making algorithms, such as game theory [9] and decision trees [10], to select maneuvers for aerial confrontation from a predefined set of actions. Another line of research designs complex decision logic based on expert experience, dynamically selecting appropriate tactics or behaviors in response to situational changes. For example, Hou et al. [11] propose a decision-making framework based on the State-Event-Condition-Action (SECA) methodology, which captures typical events in aerial confrontation to enable tactical adaptation and autonomous decision-making. Other research integrates the SECA framework with game theory to enhance the effectiveness of aerial confrontation

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[12]. However, rule-based approaches have two main limitations: (i) Human expertise may be incomplete, which easily leads to constructed rules failing to cover all possible situations or causing rule conflicts. Furthermore, the complexity of rule construction increases significantly with the scale of the rule set. (ii) Rules are often designed for specific scenarios or missions, resulting in limited generalizability across diverse operational environments.

The rapid development of Deep Reinforcement Learning (DRL) in recent years has drawn increasing research attention to DRL-based aerial confrontation strategies. These methods enable agents to autonomously learn optimal decision-making strategies through environmental interactions, demonstrating strong adaptability and self-learning capabilities. Some studies utilize enhanced DRL algorithms to develop decision-making strategies for one-on-one close-range autonomous aerial confrontation [13]. The AlphaDogfight Trials have demonstrated that AI agents can not only master basic fighter maneuvers in close-range combat but also outperform human expert pilots [14]. Other research integrates expert knowledge with DRL to improve decision-making levels and tactical diversity [15], and extends to one-on-one BVR aerial confrontation scenarios [16]. However, DRL-based methods have two main limitations: (i) The training of DRL models demands substantial data and time investments, particularly for complex tasks. This process can potentially exhibit issues such as training instability and limited policy generalization. (ii) The “black-box” nature of DRL results in poor interpretability. Although some studies have improved this by integrating expert knowledge, the fundamental issue remains unresolved, which restricts its practical value.

Research on single-fighter or close-range aerial confrontation has reached relative maturity, and current MUM-T studies mainly focus on collaborative control and combat planning [17], [18]. In contrast, there is a notable lack of research on autonomous decision-making for MUM-T BVR aerial confrontation, and a mature theoretical and methodological framework has yet to be established.

B. Motivations

The above analysis demonstrates that current rule-based and DRL-based approaches cannot fully meet the requirements of MUM-T BVR aerial confrontation. The central problem lies in how to fully utilize the respective advantages of MAVs and UAVs under different operational situations to achieve efficient human-machine collaboration. To tackle this problem, three key challenges must be addressed:

- **High complexity of MUM-T BVR aerial confrontation:** The inherent complexity of multi-fighter BVR engagement arises from multiple intertwined factors, involving situational awareness [19], threat assessment [20], weapon guidance, maneuver control, and more. The core challenge is how to construct an effective strategy that accounts for all the above elements simultaneously.

- **Difficulty in constructing deep human-machine collaborative strategies:** Given constraints in aerial resources (e.g., sensors, weapons, fuel), the core challenge is to optimize resource

utilization and achieve cooperative combat between MAVs and UAVs, thereby producing a “1+1>2” effect in overall formation effectiveness. Additionally, the strategy must be fully interpretable.

- **Complex single-fighter decision-making:** Individual fighters must dynamically adjust their maneuver control and attack decisions in response to the evolving combat environment [21], [22]. The core challenge lies in maintaining an effective offense-defense balance while actively shaping advantageous combat situations. Additionally, all decision-making processes must be fully interpretable.

C. Main Contributions

To address these challenges, this paper proposes a fully interpretable MUM-T BVR aerial confrontation framework driven by SECA rules. This framework decouples the complex problem into two layers: multi-fighter collaboration and single-fighter decision-making. At the multi-fighter level, it introduces a comprehensive mechanism for target allocation and guidance task allocation. At the single-fighter level, it designs a weapon launch strategy based on value-distance assessment and differentiated decision-making rules. Furthermore, comprehensive theoretical analysis and extensive simulation experiments collectively validate the framework’s overall effectiveness. The main contributions of this paper are threefold:

- Unlike existing studies on homogeneous multi-UAV systems [23], [24], [25], [26], this paper presents an architectural framework designed for MUM-T BVR aerial confrontation scenarios. The framework comprehensively considers various factors, including situational awareness, target allocation, tactical transition, maneuver control, threat evasion, weapon employment, and sensor guidance. To manage complexity, the framework employs a layered architecture that integrates human-machine collaboration mechanisms at the multi-fighter coordination level while leveraging platform-specific advantages at the single-fighter decision-making level.

- Unlike existing DRL-based studies [27], [28], [29], [30], this paper constructs a fully interpretable aerial confrontation framework, avoiding the “black-box” dilemma and enhancing practical utility. The framework ensures complete interpretability across all decision levels, spanning multi-fighter coordination for target and guidance allocation, as well as single-fighter maneuver and weapon employment decisions.

- Unlike existing rule-based studies [10], [11], [12], [21], this paper designs a more comprehensive set of states, events, conditions, and actions, which extends the applicability of the framework. Additionally, this paper introduces a dedicated verification mechanism. Furthermore, by incorporating coordination outcomes as influencing factors into the rule design, the framework establishes a logical connection between the two levels, thereby forming a closed-loop decision-making process.

This paper is structured as follows: Section II outlines the MUM-T BVR aerial confrontation scenario, elements, and decision-making problems. Section III details the hierarchical decision-making framework and its implementation. Section IV

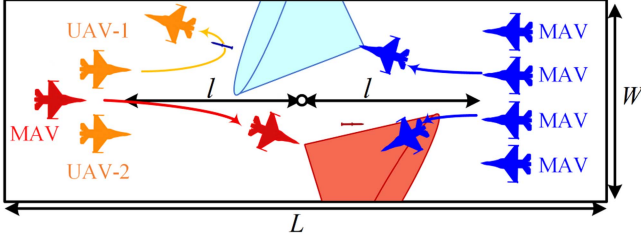


Fig. 1. Aerial confrontation scenario.

validates the framework's overall efficacy through multi-faceted simulations. Section V concludes the paper.

II. MUM-T BVR AERIAL CONFRONTATION

The objective of MUM-T BVR aerial confrontation is to achieve local air dominance. This is achieved through tight human-machine coordination to seize advantageous positions and employ weapons effectively, ultimately shooting down or driving away enemy fighters.

A. Aerial Confrontation Scenario

Fig. 1 depicts a typical MUM-T BVR aerial confrontation scenario. The red side comprises one MAV with two UAVs, opposing a blue formation of four MAVs. The engagement occurs in a free space with dimensions of L in length and W in width, with both sides initially positioned symmetrically at a distance l from the center.

Both sides start from their initial positions. When the distance is long, they use the Airborne Warning and Control System (AWACS) information to adjust flight direction. As the distance closes, they use onboard sensors and weapons to find and attack enemies, while also moving to get advantageous positions or avoid threats. The confrontation ends when one side loses all fighters, or when both sides cannot fight anymore due to a lack of fuel or missiles. Each confrontation has a maximum time limit of $\bar{T}$.

B. Aerial Confrontation Elements

This section introduces the modeling of the fighter, onboard sensor, missile, and AWACS.

- **Fighter:** This paper constructs a six-degree-of-freedom fighter model using the North-East-Down coordinate system shown in Fig. 2(a). The kinematic and dynamic models of the fighter can be defined as

$$\begin{aligned} \dot{p} &= R_b^e v^b, \\ \dot{v}^b &= \frac{f^t + f^a}{m} + (R_b^e)^T g - \omega \times v^b, \\ \dot{\Theta} &= H(\Theta)\omega, \\ \dot{\omega} &= J^{-1}(\tau - \omega \times J\omega), \end{aligned} \quad (1)$$

where, p is the fighter position, R_b^e is the rotation matrix transforming the body frame to the Earth frame, v^b is the velocity in the body frame, f^t and f^a are the thrust force and aerodynamic

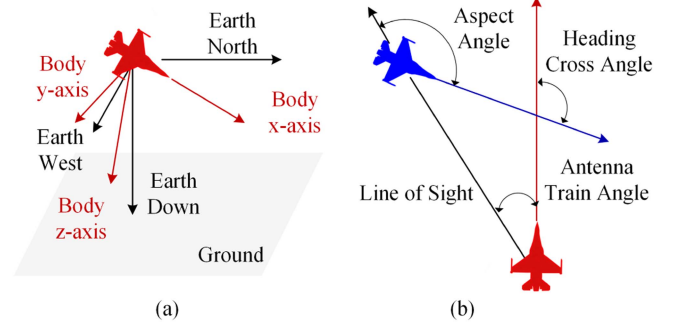


Fig. 2. Body frame and aerial confrontation angles.

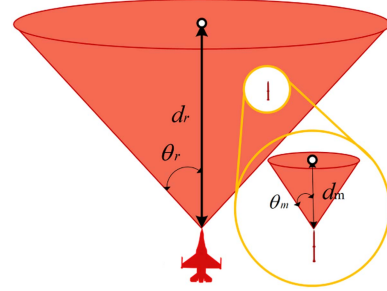


Fig. 3. Detection ranges of FCR and missile seeker.

force, m is the mass, g is the gravitational acceleration, ω is the angular velocity, $\Theta = [\phi, \theta, \psi]^T$ are the Euler angles, $H(\Theta)$ is the matrix transforming ω to the rates of the Euler angles, J is the inertia tensor, and τ is the moment.

This paper uses line of sight (LOS), distance, antenna tracking angle (ATA), azimuth angle (AA), and heading cross angle (HCA) to describe the positional relationship between fighters, as shown in Fig. 2(b), which can be defined as

$$\begin{aligned} d &= \|LOS\|_2, \\ ATA &= \arccos(LOS^T h/d), \\ AA &= \arccos(LOS^T h^e/d), \\ HCA &= \arccos(h^T h^e), \end{aligned} \quad (2)$$

where, d is the distance between the two fighters, h is the heading angle of one fighter, and h^e is the heading angle of the other fighter.

This paper assumes that a fighter is considered to have crashed when its altitude is less than $\underline{h}$. If the distance between two fighters is less than $\underline{d}$, they are considered to have collided.

Airborne sensor: The onboard sensors considered in this paper comprise fire-control radar (FCR), radar warning receivers (RWR), and missile warning systems (MWS), which provide pilots with situational awareness and threat alerts.

The FCR model is simplified as a conical region with a specific detection range $d_r \leq \bar{d}^{\text{radar}}$ and an angle $|\theta_r| \leq \bar{\theta}^{\text{radar}}$, as shown in Fig. 3. The radar's central axis remains aligned with the fighter's orientation without any bias error. The FCR operates continuously throughout aerial confrontation to obtain information on enemy fighters within its detection region and to

provide guidance support for missiles. However, due to ground clutter interference, the FCR cannot detect enemy fighters in the downward-looking direction with a relative radial velocity $V_r \geq \bar{V}^{\text{radial}}$ to own-side fighter.

When enemy FCR illuminates and locks onto own-side fighter, the RWR issues a radar lock-on alert, which usually indicates that the enemy has launched a missile. When the enemy missile seeker locks onto own-side fighter, the MWS continuously provides directional and distance data to support evasive maneuvers.

Missile: After launch, the missile enters midcourse guidance. The fighter provides target updates via a fighter-to-missile data link to maintain the missile's track. If the link is interrupted, the missile continues tracking using the last known target data, assuming constant-speed linear target motion. When the target enters the seeker's detection range, the missile switches to terminal guidance and tracks the target using seeker data.

The radar seeker model is simplified as a conical region with a detection range $d_m \leq \bar{d}^{\text{missile}}$ and an angle $|\theta_m| \leq \bar{\theta}^{\text{missile}}$, as shown in Fig. 3. The missile uses proportional navigation guidance. The missile's effective kill range is $\bar{d}^{\text{kill}}$, and target destruction is confirmed upon impact within $\bar{d}^{\text{kill}}$.

AWACS: AWACS provides situational awareness support by updating enemy fighter positions and velocities at frequency f_A , with corresponding errors of $\pm\sigma_p$ and $\pm\sigma_v$.

C. Aerial Confrontation Decision-Making Problem

The core of the decision-making problem lies in establishing the mapping relationship from observations to control inputs. This section formally describes observations, control inputs, and the decision function.

For either side, the complete observation set $s(t)$ obtainable by a single fighter is defined as

$$s(t) = \{s_{\text{ff}}(t), s_{\text{ef}}(t), s_{\text{fm}}(t), s_{\text{em}}(t)\}, \quad (3)$$

where, $s_{\text{ff}}(t)$, $s_{\text{ef}}(t)$, $s_{\text{fm}}(t)$, and $s_{\text{em}}(t)$ are the observed information of friendly fighters, enemy fighters, friendly missiles, and enemy missiles, respectively. Let the number of friendly fighters be M , each denoted as $F_i (i = 1, 2, \dots, M)$. Let the number of enemy fighters be N , each denoted as $F_j (j = 1, 2, \dots, N)$. Each fighter carries K missiles, with friendly and enemy missiles denoted as $M_{ik} (i = 1, 2, \dots, M; k = 1, 2, \dots, K)$ and $M_n (n = 1, 2, \dots, NK)$, respectively.

This paper adopts the same aerial confrontation environment as [5], [12], [16], [21], [30]. The information of each friendly fighter $s_{\text{ff}}^i(t)$ includes the fighter identification number ID_i , position $p_i = [x_i, y_i, z_i]$, attitude $\Theta_i = [\psi_i, \theta_i, \phi_i]$, velocity $v_i = [v_{i,x}, v_{i,y}, v_{i,z}]$, fuel F , remaining missile count m_{left} , radar warning status λ_R , and missile warning status λ_M . Warning statuses are boolean variables indicating active warning conditions. Specifically, it is defined as

$$s_{\text{ff}}^i(t) = [\text{ID}_i, p_i, \Theta_i, v_i, F, m_{\text{left}}, \lambda_R, \lambda_M]. \quad (4)$$

Using the FCR and AWACS, the friendly side obtains information $s_{\text{ef}}^j(t)$ for each enemy fighter, which is defined as

$$\begin{aligned} s_{\text{radar}}^j(t) &= [\text{ID}_j, p_j, v_j], \\ s_{\text{awacs}}^j(t) &= [\text{ID}_j, p_{j,\sigma}, v_{j,\sigma}], \\ s_{\text{ef}}^j(t) &= \max [s_{\text{radar}}^j(t), s_{\text{awacs}}^j(t)], \end{aligned} \quad (5)$$

where ID_j , $p_j = [x_j, y_j, z_j]$, and $v_j = [v_{j,x}, v_{j,y}, v_{j,z}]$ are the identification number, position, and velocity of the enemy fighter, respectively. $p_{j,\sigma}$ and $v_{j,\sigma}$ are the corresponding values with errors. Since both FCR data $s_{\text{radar}}^j(t)$ and AWACS data $s_{\text{awacs}}^j(t)$ may suffer from interruptions or repetitions, information fusion from these two sources is required.

The information of each friendly missile $s_{\text{fm}}^{ik}(t)$ includes the identification number of the launching fighter ID_i , the missile identification number ID_{ik} , the target identification number ID_j , the guidance source identification number ID_{guide} , the position $p_{ik} = [x_{ik}, y_{ik}, z_{ik}]$, the attitude $\Theta_{ik} = [\psi_{ik}, \theta_{ik}, \phi_{ik}]$, the velocity $v_{ik} = [v_{ik,x}, v_{ik,y}, v_{ik,z}]$, and the terminal guidance flag λ_T . λ_T is a boolean variable indicating activation of the terminal guidance phase. Specifically, it is defined as

$$s_{\text{fm}}^{ik}(t) = [\text{ID}_i, \text{ID}_{ik}, \text{ID}_j, \text{ID}_{\text{guide}}, p_{ik}, \Theta_{ik}, v_{ik}, \lambda_T]. \quad (6)$$

Using the MWS, the alerted fighter obtains information $s_{\text{em}}^{in}(t)$ on enemy missiles, which is defined as

$$s_{\text{em}}^{in}(t) = [d_{in}, \theta_{in}, \varphi_{in}], \quad (7)$$

where d_{in} , θ_{in} , and φ_{in} are the distance, azimuth, and elevation of enemy missile n relative to friendly fighter i , respectively.

The mapping function χ from the observation $s(t)$ to the control input $u(t)$ is denoted as

$$u(t) = \chi(s(t)), \quad (8)$$

where, the control inputs $u(t)$ comprise the load factor n_y , roll angle ϕ , throttle setting T , and a boolean variable λ_m for missile launch authorization.

III. HIERARCHICAL DECISION-MAKING FRAMEWORK

A. Overall Framework

The SECA method integrates the event-triggering mechanism of Event-Condition-Action (ECA) [31] with the state-transition logic of Finite State Machines (FSM) [32]. This enables the system to efficiently handle events in complex dynamic environments while maintaining clear and scalable decision logic.

The SECA model is a six-tuple $\mathcal{M} = (ID, S, E, C, A, \Pi)$, where $S = \{s_1, s_2, \dots, s_{m_1}\}$, $E = \{e_1, e_2, \dots, e_{m_2}\}$, $C = \{c_1, c_2, \dots, c_{m_3}\}$, and $A = \{a_1, a_2, \dots, a_{m_4}\}$ are the finite sets of states, events, conditions, and actions, respectively. Π is the state transition logic that dynamically determines the next action and state based on the current state, event, and condition, expressed as $(a, s') = \Pi(s, e, c)$. ID is a unique serial number. Π comprises a set of predefined decision rules that govern the entity's dynamic behavior across varying situations. Fig. 4 shows an example of a simple SECA rule in an aerial confrontation scenario.

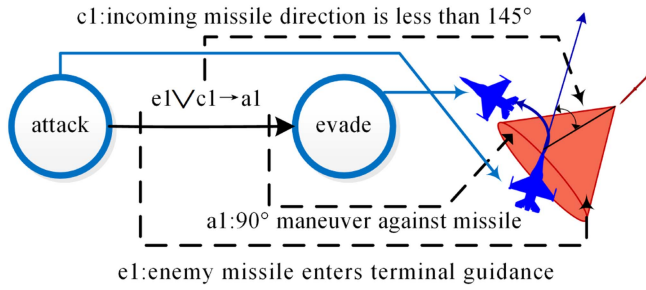


Fig. 4. Example of a SECA rule in an aerial confrontation scenario. The fighter begins in the attack state. Upon triggering the event of an enemy missile entering terminal guidance and satisfying the condition that the incoming missile direction is less than 145° , the fighter executes a 90° maneuver against the missile and transitions to the evade state.

Based on the SECA method, this paper proposes a hierarchical decision-making framework. The framework consists of four modules: multi-fighter coordination, single-fighter decision-making, low-level control, and environment interaction, as shown in Fig. 5.

In the multi-fighter coordination layer, the formation processes real-time situational information $s(t)$ to perform target assignment and missile guidance allocation. In the single-fighter decision-making layer, each fighter analyzes $s(t)$ to determine maneuvering and weapon employment strategies according to assigned tasks. In the low-level control layer, control commands $u(t)$ are computed based on selected strategies and applied to the fighter model. In the environment interaction layer, the confrontation is advanced, and the updated observations $s(t)$ are fed back to the multifighter coordination layer and the singlefighter decisionmaking layer. Even under identical situations, different targets, guidance tasks, and platform types can each lead to different strategy selections.

B. Multi-Fighter Coordination Layer

During aerial confrontation, the number of fighters, relative positional relationships, and combat status change continuously for both sides. Simple target allocation rules are insufficient to cope with such complexity. Therefore, this paper proposes a comprehensive target allocation mechanism that organically integrates global assignment with supplementary responses to localized emergency threats.

Based on situational information from fighters on both sides, this paper selects friendly fighters that are on the offensive to participate in global target allocation. The Hungarian algorithm assigns friendly fighters to enemy fighters using a weight function defined as

$$W_{ij}^t = G_t(d) \cdot G_t(\psi) \cdot G_t(F) \cdot G_t(S) \cdot G_t(R) \cdot G_t(T), \quad (9)$$

where, W_{ij}^t is the target matching weight between the friendly fighter F_i and the enemy fighter F_j . $G_t(d)$, $G_t(\psi)$, $G_t(F)$, $G_t(S)$, $G_t(R)$, and $G_t(T)$ are the scoring functions for distance, heading angle, fuel, status, FCR detection, and target stability, respectively.

The specific scoring functions are defined as

$$G_t(d) = \begin{cases} 0.1 & \text{if } d < 20 \text{ km} \\ 1 - 0.9e^{-0.2(d-20)} & \text{if } 20 \leq d < 40 \text{ km} \\ 1 & \text{if } 40 \leq d \leq 60 \text{ km} \\ 1 + 0.7 \ln\left(1 + \frac{d-60}{60}\right) & \text{if } d > 60 \text{ km} \end{cases}$$

$$G_t(\psi) = \begin{cases} 1 & \text{if } \psi \leq 30^\circ \\ 1 + 0.5 \ln\left(1 + \frac{\psi-30}{150}\right) & \text{if } 30 < \psi \leq 180^\circ \end{cases}$$

$$G_t(F) = \begin{cases} 2 & \text{if } F < 1500 \text{ lb} \\ 1.5 + 0.5e^{-\frac{F-1500}{1500}} & \text{if } 1500 \leq F \leq 3000 \text{ lb} \\ 1 & \text{if } F > 3000 \text{ lb} \end{cases}$$

$$G_t(S) = \begin{cases} 1 & \text{if } S = S_{\text{attack}} \\ 1.2 & \text{if } S = S_{\text{defensive attack}} \end{cases}$$

$$G_t(R) = \begin{cases} 1 & \text{if } R = R_{\text{detected}} \\ 1.2 & \text{if } R = R_{\text{not detected}} \end{cases}$$

$$G_t(T) = \begin{cases} 1.5 & \text{if } T_{\text{last}} \text{ exists and } T \neq T_{\text{last}} \\ 1 & \text{otherwise} \end{cases}, \quad (10)$$

where, S_{attack} is the offensive state with full combat capability and no weapon threats. $S_{\text{defensive attack}}$ is an offensive state with defensive considerations when locked by enemy radar, carrying a higher security risk and thus a higher allocation cost. T is the enemy target identified at the current moment, and T_{last} is the target assigned previously. Target changes require abandoning the original mission, thereby incurring a higher cost to maintain mission stability. R_{detected} means the FCR has detected the current target. $R_{\text{not detected}}$ means no detection, which reduces initial combat effectiveness and thereby incurs a higher allocation cost.

Remark 1: Unless otherwise specified, the units are unified as follows: distance in kilometers (km), angles in degrees ($^\circ$), fuel quantity in pounds (lb), speed in Mach units (Mach), time in seconds (s), and altitude in meters (m).

The global target allocation preferentially assigns enemy fighters to friendly fighters meeting key criteria, including appropriate range (40-60 km), small yaw angle ($< 30^\circ$), sufficient fuel (> 1500 lb), low threat level, radar detectability, and track stability. This design can create conditions for weapon launch while reducing the threat to itself. The result of target allocation will serve as an input to influence fighter decisions on maneuvering and weapon employment.

The global target allocation prioritizes one-to-one assignments. The excess friendly fighters are reassigned to create local superiority when in a numerical advantage. Conversely, when at a numerical disadvantage, the friendly side proactively selects higher-scoring targets.

To effectively address emergencies, this paper designs a supplementary allocation for local emergency threats based on the global target allocation, enhancing the flexibility and adaptability of the allocation process. This paper identifies three typical emergency scenarios:

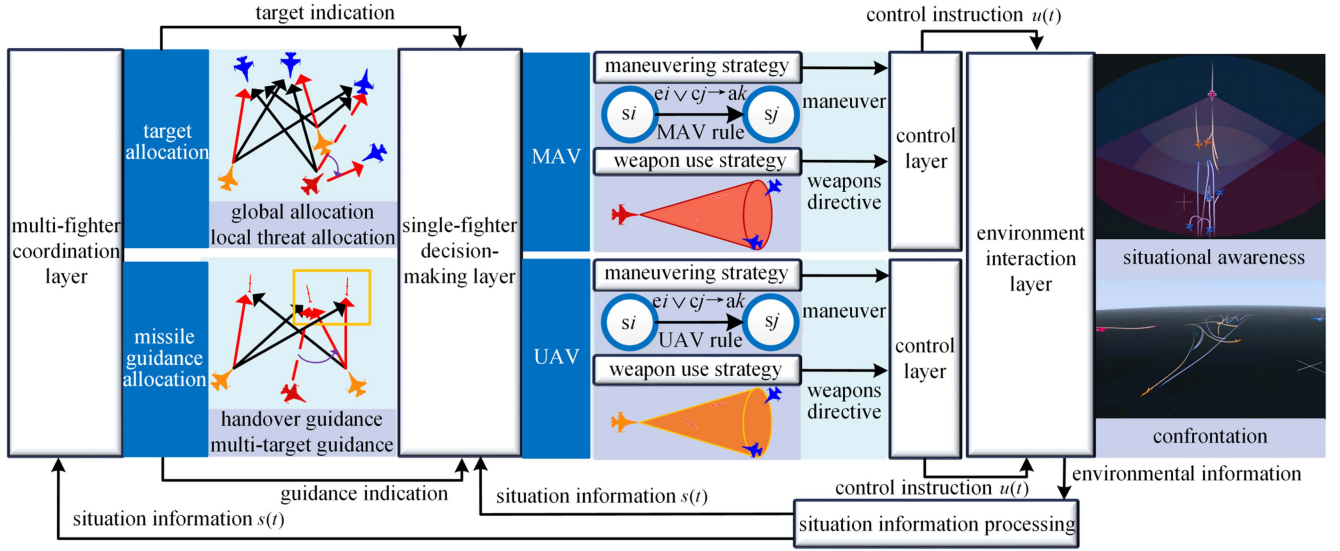


Fig. 5. The hierarchical decision-making framework.

- If a high-threat target emerges at close range and the originally assigned target is located at a distance exceeding this close range by 20 km or more, the fighter engages the closer target.
- If a threat target emerges at close range and the AA of the originally assigned target is less than 60° in absolute value, the fighter engages the closer target.
- If other high-threat or threat targets emerge at close range but do not satisfy the above conditions, the fighter maintains the current target but is allowed to launch missiles at the new target when appropriate.

Remark 2: This paper defines threat targets as enemy fighters that satisfy the basic conditions for launching weapons against own-side fighter. Those located within own-side fighter's Minimum Abort Range are classified as high-threat targets.

This supplementary allocation mechanism enables fighters to respond to emerging threats or to effectively utilize weapons to reduce the overall threat level.

Besides target assignment, guidance task allocation is another major challenge in cooperative engagement. With the dynamic changes in missile guidance requirements, relative positions, and combat status, simple sensor resource allocation rules fail to leverage the full potential of the sensor group. Therefore, this paper proposes a comprehensive guidance task allocation mechanism that organically integrates global assignment with optimization for multi-target guidance.

Based on the situational information of the friendly fighters and missiles, this paper selects missiles in the mid-course guidance phase that lack stable guidance sources, as well as friendly fighters that are on offensive or auxiliary, to participate in the global guidance task allocation. The Hungarian algorithm assigns friendly fighters to the selected missiles using a weight function defined as

$$W_{i'k}^g = G_g(d) \cdot G_g(\psi) \cdot G_g(F) \cdot G_g(S) \cdot G_g(R) \cdot G_g(d_m), \quad (11)$$

where, $W_{i'k}^g$ is the guidance allocation weight between the fighter F_i and the missile $M_{i'k}$. $G_g(d)$, $G_g(\psi)$, $G_g(F)$, $G_g(S)$,

$G_g(R)$, and $G_g(d_m)$ are the scoring functions for the fighter-target distance, fighter-target yaw angle, fuel, status, FCR detection, and fighter-missile distance, respectively.

The specific scoring functions are defined as

$$G_g(d) = \begin{cases} 1.5 - 0.5e^{-0.1(40-d)} & \text{if } d < 40 \text{ km} \\ 1 & \text{if } 40 \leq d \leq 90 \text{ km} \\ 1.5 - 0.5e^{-0.1(d-90)} & \text{if } 90 < d \leq 100 \text{ km} \\ 2 & \text{if } d > 100 \text{ km} \end{cases},$$

$$G_g(\psi) = \begin{cases} 1 & \text{if } 0 \leq \psi \leq 45^\circ \\ 1.2 - 0.2e^{-0.1(\psi-45)} & \text{if } 45 < \psi \leq 60^\circ \\ 1.2 + 0.8 \frac{\psi-60}{60} & \text{if } 60 < \psi \leq 120^\circ \\ 2 & \text{if } 120 < \psi \leq 180^\circ \end{cases},$$

$$G_g(F) = \begin{cases} 2 & \text{if } F < 1500 \text{ lb} \\ 1.5 + 0.5e^{-\frac{F-1500}{1500}} & \text{if } 1500 \leq F \leq 3000 \text{ lb} \\ 1 & \text{if } F > 3000 \text{ lb} \end{cases},$$

$$G_g(S) = \begin{cases} 1 & \text{if } S = S_{\text{attack}} \text{ and } S \neq S_{\text{guidance}} \\ 1.2 & \text{if } S = S_{\text{defensive attack}} \text{ and } S \neq S_{\text{guidance}} \\ 0.2 & \text{if } S = S_{\text{auxiliary}} \text{ and } S \neq S_{\text{guidance}} \\ 0.9 & \text{if } S = S_{\text{guidance}} \text{ and target is consistent} \\ 1.5 & \text{if } S = S_{\text{guidance}} \text{ and target is inconsistent} \end{cases},$$

$$G_g(R) = \begin{cases} 1 & \text{if } R = R_{\text{detected}} \\ 1.2 & \text{if } R = R_{\text{not detected}} \end{cases},$$

$$G_g(d_m) = \begin{cases} 1 & \text{if } d_m < 50 \text{ km} \\ 1.5 - 0.5e^{-0.01(d_m-50)} & \text{if } 50 \leq d_m \leq 300 \text{ km} \\ 2 & \text{if } d_m > 300 \text{ km} \end{cases} \quad (12)$$

where, $S_{\text{auxiliary}}$ indicates that the fighter has no missiles remaining and cannot attack directly. However, its sensors are still available to support other friendly fighters. A lower cost is set for this state to make full use of the missile-less fighter's

capabilities, allowing other friendly fighters to perform other tactical tasks more flexibly. S_{guidance} means the fighter has been assigned a guidance task, which is discussed in two cases. When the target is consistent, indicating minimal disruption to the current guidance task, a relatively low cost is applied. This facilitates the full utilization of the fighter's multi-target guidance capability. When the target is inconsistent, indicating a conflict in the guidance target, a higher cost is set to avoid disrupting the original guidance task and to minimize unnecessary target switching.

The global guidance task allocation preferentially assigns missiles to friendly fighters meeting key criteria, including appropriate target distance (40-90 km), moderate yaw angle ($< 45^\circ$), sufficient fuel (> 1500 lb), low threat level, radar detectability, and appropriate missile distance (< 50 km). This design can create conditions for missile guidance while reducing the threat to the fighters and fully utilizing the remaining value of missile-less fighters. Fighters in the auxiliary state are prioritized for guidance tasks. If assigned a task, the auxiliary fighter locks onto that guidance target. Otherwise, it locks onto the target of the nearest friendly fighter. The result of guidance allocation will serve as an input to influence the fighters' maneuvering and weapon employment decisions.

When the original guidance source cannot provide guidance information, this mechanism can dynamically adjust sensor resources to reassign missile guidance sources, enabling alternate or relay guidance and extending the total guidance time. The guidance task is terminated when the missile enters the terminal guidance phase or exhausts its energy.

Missile guidance is not a mandatory task. Fighters should not forgo more important tasks or put themselves in danger to complete guidance missions. Therefore, this paper designs a threshold $\bar{W}^g = 2.5$. When the guidance allocation weight exceeds this threshold, the fighter will abandon the current guidance task and switch to more important missions.

With the continuous improvement of FCR performance, modern fighters are now capable of multi-target guidance, allowing them to guide multiple missiles to attack several targets simultaneously. Therefore, this paper designs a multi-target guidance optimization allocation mechanism. Fighters that have already undertaken guidance tasks can provide guidance information for other missiles on targets within their radar detection range, thereby enhancing the efficiency of sensor utilization. This leads to a change in the original guidance task allocation.

C. Single-Fighter Decision-Making Layer

During aerial confrontation, each fighter operates in a highly complex and dynamic environment with a limited missile load-out. Consequently, simple missile launch strategies are insufficient for enhancing missile utilization efficiency. Therefore, this paper designs a missile launch strategy based on value-distance judgment. The fighter will autonomously launch the missile when the launch conditions are met. The value-distance can be defined as

$$d_{\text{value}} = G_{\text{fire}} \cdot G_{\text{left}} \cdot G_{\text{track}} \cdot \bar{d}_{\text{fire}}, \quad (13)$$

where, G_{fire} is the basic launch condition score, evaluating whether the missile satisfies the range and off-boresight angle constraints. G_{left} is the score based on the number of remaining missiles. G_{track} is the score based on the number of missiles currently tracking the target. $\bar{d}_{\text{fire}}$ is the maximum launch distance. Specifically, these are defined as

$$G_{\text{fire}} = \begin{cases} \min\left(1, 3 - \frac{20d}{9\bar{d}_{\text{fire}}}\right) & \text{if } |\theta_e| \leq \bar{\theta}_{\text{fire}} \text{ and } |\psi_e| \leq \bar{\psi}_{\text{fire}} \text{ and } |d| \leq \bar{d}_{\text{fire}}, \\ 0 & \text{otherwise} \end{cases}$$

$$G_{\text{left}} = \frac{m_{\text{left}} + k - 2}{2k - 2},$$

$$G_{\text{track}} = \begin{cases} \min\left(1, 1 - \frac{1}{2}m_t\right) & \text{if } d \geq 60 \text{ km} \\ \min\left(1, 1.2 - \frac{2}{5}m_t + \frac{S_{\text{danger}}}{4}\right) & \text{if } 20 \leq d < 60 \text{ km}, \\ \min\left(1, 1 - \frac{1}{2}m_t + S_{\text{danger}}\right) & \text{if } d < 20 \text{ km} \end{cases} \quad (14)$$

where, θ_e , ψ_e , d , $\bar{\theta}_{\text{fire}}$, $\bar{\psi}_{\text{fire}}$, $\bar{d}_{\text{fire}}$, m_{left} , m_{track} , and S_{danger} are the expected pitch angle, expected yaw angle, actual distance to the target, maximum launch pitch angle, maximum launch yaw angle, maximum launch distance, the number of remaining missiles on the fighter, the number of missiles currently tracking the target, and the fighter's threat state flag, respectively. If the fighter is under any weapon or security threats, S_{danger} is set to 1. Otherwise, it is set to 0. The fewer missiles remain, the stricter the launch conditions to ensure the fighter's sustained combat capability. The more missiles are tracking the target, the stricter the launch conditions to avoid waste. Moreover, launches near the performance boundary are reduced.

The missile is deemed launch-worthy when the value-distance is no less than the actual range to the target, which is defined as $d_{\text{value}} \geq d$. When the distance between the two sides is relatively large, the fighter can adopt a high-lob trajectory to increase the missile's range. This paper directly uses empirical values to adjust the high-lob pitch angle, which is defined as $\theta_e = \min(\max(\theta_e, 0.405d), 30)$.

In aerial confrontation, certain maneuvers must be coordinated with missile launches. For example, to prevent rear pursuit by enemy fighters, a friendly fighter may launch a missile before executing a close-range turn. In such a case, the missile is primarily used for deterrence to disrupt the enemy's strategy. Accordingly, this paper designs a tactical missile launch strategy. Independent launch criteria are established for certain engagement distances, allowing a missile launch whenever $G_{\text{fire}} \neq 0$ and $m_t \leq 2$.

In MUMT, MAVs and UAVs exhibit different operational tendencies. MAVs tend to adopt conservative strategies to ensure stable command and control, while UAVs tend to employ aggressive strategies to enhance the formation's offensive capabilities. A single decision-making strategy is insufficient to meet these differentiated combat requirements.

Therefore, this paper proposes a SECA-based decision-making strategy that integrates general and specific rules. The general rules form a baseline for all fighters, with specific rules added for different operational tendencies. This design reduces the workload of developing comprehensive strategies for each

TABLE I
STATE SETS

Label	State	Label	State	Label	State
s1	cruise	s4	auxiliary	s7	protect
s2	attack	s5	evade	s8	guidance
s3	defensive attack	s6	defense	s9	fire

TABLE II
EVENT SETS

Label	Event
e1	AWACS detects enemy fighter
e2	FCR detects enemy fighter
e3	assigned to guidance task
e4	RWR shows lock-on
e5	RWR warning interval exceeds 90 seconds
e6	enemy missile enters terminal guidance
e7	failed to evade missile within 15 seconds
e8	successfully evade all missiles
e9	end of safety assessment time
e10	all enemy fighters destroyed
e11	fighter altitude below required lower limit
e12	fighter altitude above required upper limit
e13	fighter speed below minimum speed
e14	fighter fuel below minimum required level

TABLE III
CONDITION SETS

Label	Condition
c1	offensive posture
c2	defensive posture
c3	tail-chase posture
c4	beyond Phase I distance
c5	between Phase I distance and Phase II distance
c6	between Phase II distance and Phase III distance
c7	between Phase III distance and Phase IV distance
c8	within minimum distance
c9	missiles remaining
c10	within safety assessment time
c11	incoming angle of missile is less than 145°
c12	meet conditions for altitude reduction
c13	meet conditions for auxiliary

fighter type individually and enhances the flexibility of the whole framework.

The sets of states, events, conditions, and actions are defined in Tables I, II, III, and IV, respectively. They are denoted as L_S , L_E , L_C , and L_A . The state set L_S is divided into two categories: tactical states, which are related to fighter maneuvers, and payload states, which comprise the sensor guidance state (s_8) and weapon launch state (s_9). A fighter can be in only one tactical state at a time, but it can simultaneously be in one or more payload states.

The general SECA rules, denoted as R_G , are shown in Table V. The specific rules for MAVs and UAVs, denoted as R_{SM} and R_{SU} respectively, are shown in Tables VI and VII. Fig. 6 presents the state transition diagram, where black, red, and orange lines correspond to general rules, special rules for MAVs, and special rules for UAVs, respectively. These specific rules primarily take into account the engagement line distance, the posture of both sides, and the requirements of the guidance mission.

TABLE IV
ACTION SETS

Label	Action
a0	tactical launch missile
a1	circle
a2	track
a3	right 90° maneuver against missile
a4	left 90° maneuver against missile
a5	turn maneuver against missile
a6	right 30° offset against fighter
a7	left 30° offset against fighter
a8	right 45° offset against fighter
a9	left 45° offset against fighter
a10	turn maneuver against fighter
a11	emergency right offset against fighter with altitude change
a12	emergency left offset against fighter with altitude change
a13	climb
a14	descend
a15	accelerate
a16	fly straight
a17	return to base

TABLE V
GENERAL SECA RULES

No.	S	E	C	A	S'
1	sall	e12	-	a14	s7
2	sall	$\neg e12 \wedge e13$	-	a15	s7
3	sall	$e11 \wedge \neg e13$	-	a13	s7
4	sall	$\neg e11 \wedge \neg e12 \wedge \neg e13$ $\wedge \neg e6 \wedge e14$	-	a17	s7
5	s1-4,7	e6	c11	a3/4	s5
6	s1-4,7	e6	$\neg c11$	a5	s6
7	s2-4	e10	-	a17	s1
8	s1	-	-	a1	s1
9	s1	$e1 \vee e2$	-	a2	s2
10	s1	e4	-	-	s3
11	s2	-	c4	a2	s2
12	s2	e4	-	-	s3
13	s2	-	$\neg c9$	-	s4
14	s3	$e5 \wedge \neg e10$	-	a2	s2
15	s3	$\neg e5$	c4	a2	s3
16	s3	-	$\neg c9$	-	s4
17	s4	$\neg e3 \wedge \neg e10$	$\neg c13$	a17	s4
18	s4	e3	$\neg c13$	a8/9	s4
19	s4	-	$c4 \wedge c13$	a2	s4
20	s5	$\neg e6 \wedge e9$	-	-	s2
21	s5	e6	c8	$(a0 \vee \neg a0)$ $\wedge a3/4$	s5
22	s5	$\neg e6$	c10	a16	s5
23	s5	e6	$\neg c8$	a3/4	s5
24	s5	$e7 \wedge \neg e10$	-	a5	s6
25	s5	$e6 \wedge \neg e10$	c10	a5	s6
26	s5	$e6 \wedge e10$	-	a5	s6
27	s6	$\neg e6 \wedge e9$	-	-	s2
28	s6	e6	c8	$(a0 \vee \neg a0)$ $\wedge a5$	s6
29	s6	$\neg e6$	c10	a16	s6
30	s6	e6	$\neg c8$	a5	s6
31	s7	$\neg e11 \wedge \neg e12 \wedge$ $\neg e13 \wedge \neg e14$	-	-	s2

Additionally, whether to execute tactical missile launches $a0$ will directly impact the selection of subsequent maneuvering actions, as shown in rule 34. The differentiated rule design fully considers the combat characteristics and mission requirements of the two fighter types.

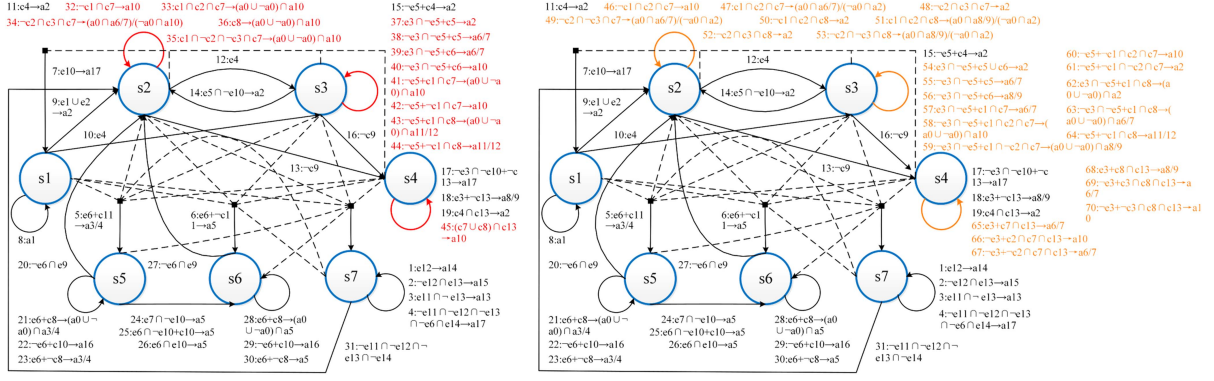


Fig. 6. SECA rules state transition diagram. (a) MAV. (b) UAV. Dashed lines serve to mark rules that are valid for multiple states.

TABLE VI
MAV SPECIAL SECA RULES

No.	S	E	C	A	S'
32	s2	-	$\neg c1 \wedge c7$	a10	s2
33	s2	-	$c1 \wedge c2 \wedge c7$	$(a0 \vee \neg a0) \wedge a10$	s2
34	s2	-	$\neg c2 \wedge c3 \wedge c7$	$(a0 \wedge a6/a7) / (\neg a0 \wedge a10)$	s2
35	s2	-	$c1 \wedge \neg c2$	$(a0 \vee \neg a0) \wedge a10$	s2
36	s2	-	$\neg c3 \wedge c7$	$(a0 \vee \neg a0) \wedge a10$	s2
37	s3	$e3 \wedge \neg e5$	c5	a2	s3
38	s3	$\neg e3 \wedge \neg e5$	c5	a6/7	s3
39	s3	$e3 \wedge \neg e5$	c6	a6/7	s3
40	s3	$\neg e3 \wedge \neg e5$	c6	a10	s3
41	s3	$\neg e5$	$c1 \wedge c7$	$(a0 \vee \neg a0) \wedge a10$	s3
42	s3	$\neg e5$	$\neg c1 \wedge c7$	a10	s3
43	s3	$\neg e5$	$c1 \wedge c8$	$(a0 \vee \neg a0) \wedge a11/a12$	s3
44	s3	$\neg e5$	$\neg c1 \wedge c8$	a11/a12	s3
45	s4	-	$(c7 \vee c8) \wedge c13$	a10	s4

TABLE VII
UAV SPECIAL SECA RULES

No.	S	E	C	A	S'
46	s2	-	$\neg c1 \wedge c2 \wedge c7$	a10	s2
47	s2	-	$c1 \wedge c2 \wedge c7$	$(a0 \wedge a6/a7) / (\neg a0 \wedge a2)$	s2
48	s2	-	$\neg c2 \wedge c3 \wedge c7$	a2	s2
49	s2	-	$\neg c2 \wedge \neg c3 \wedge c7$	$(a0 \wedge a6/a7) / (\neg a0 \wedge a2)$	s2
50	s2	-	$\neg c1 \wedge c2 \wedge c8$	a2	s2
51	s2	-	$c1 \wedge c2 \wedge c8$	$(a0 \wedge a8/a9) / (\neg a0 \wedge a2)$	s2
52	s2	-	$\neg c2 \wedge c3 \wedge c8$	a2	s2
53	s2	-	$\neg c2 \wedge \neg c3 \wedge c8$	$(a0 \wedge a8/a9) / (\neg a0 \wedge a2)$	s2
54	s3	$e3 \wedge \neg e5$	$c5 \vee c6$	a2	s3
55	s3	$\neg e3 \wedge \neg e5$	c5	a6/7	s3
56	s3	$\neg e3 \wedge \neg e5$	c6	a8/9	s3
57	s3	$e3 \wedge \neg e5$	$c1 \wedge c7$	a6/7	s3
58	s3	$\neg e3 \wedge \neg e5$	$c1 \wedge c2 \wedge c7$	$(a0 \vee \neg a0) \wedge a10$	s3
59	s3	$\neg e3 \wedge \neg e5$	$c1 \wedge \neg c2 \wedge c7$	$(a0 \vee \neg a0) \wedge a8/9$	s3
60	s3	$\neg e5$	$\neg c1 \wedge c2 \wedge c7$	a10	s3
61	s3	$\neg e5$	$\neg c1 \wedge \neg c2 \wedge c7$	a2	s3
62	s3	$e3 \wedge \neg e5$	$c1 \wedge c8$	$(a0 \vee \neg a0) \wedge a2$	s3
63	s3	$\neg e3 \wedge \neg e5$	$c1 \wedge c8$	$(a0 \vee \neg a0) \wedge a6/7$	s3
64	s3	$\neg e5$	$\neg c1 \wedge c8$	a11/12	s3
65	s4	e3	$c7 \wedge c13$	a6/7	s4
66	s4	$\neg e3$	$c2 \wedge c7 \wedge c13$	a10	s4
67	s4	$\neg e3$	$\neg c2 \wedge c7 \wedge c13$	a6/7	s4
68	s4	e3	$c8 \wedge c13$	a8/9	s4
69	s4	$\neg e3$	$c3 \wedge c8 \wedge c13$	a6/7	s4
70	s4	$\neg e3$	$\neg c3 \wedge c8 \wedge c13$	a10	s4

In addition to the states previously defined, the fighter employs evade and defense states to handle threats. The evade state involves a lower degree of avoidance, enabling a swift return to offensive actions. In contrast, the defense state entails stronger avoidance maneuvers to evade more serious threats. The protect

state is a corrective mode triggered when the fighter enters an unsafe condition. This condition is defined as a violation of any of the following constraints: altitude outside the prescribed range $[H_{lower}, H_{upper}]$, speed below V_{min} , or fuel falling below F_{min} . The safety assessment time T_{safe} is defined as the mandatory threat assessment period after disengaging from all threats. T_{safe} is reset when $e8$ is triggered, and the fighter may only transition to an offensive state once T_{safe} has ended. $c12$ requires the altitude to be no less than $6H_{lower}$ and allows further descent during $a3$ – $a5$. $c13$ is satisfied when at least one enemy fighter is present, and at least one friendly fighter with missiles remains. The choice of left or right offset maneuver depends on the specific relative positional relationship.

The increasing number of considerations significantly complicates the design of mutually exclusive rules. Existing approaches often rely on global predefined priorities [11] or introduce forced classifications [12] to prevent simultaneous rule triggering. Although these methods simplify execution logic, they either increase reliance on expert knowledge or introduce new decision-making problems. This paper proposes a state-based rule execution mechanism. When multiple rules are triggered simultaneously, the system prioritizes those leading to higher threat states ($s1 \rightarrow s7$). This approach facilitates clearer logic organization during design, reduces complexity, and avoids introducing new issues.

The execution logic of the hierarchical decision-making framework is shown in Algorithm 1.

D. Rule Correctness Analysis

Based on the SECA methodology [11], this paper proposes a verification method for rule correctness, covering both feasibility and effectiveness.

Rule feasibility requires state transitions to execute properly under the following constraints: (i) States are mutually exclusive and each has dedicated entry and exit rules; (ii) Only one rule can be executed in a state at the same time, and each rule has the potential to be triggered; (iii) Multiple simultaneous transitions are permitted but must terminate within finite steps. For large rule sets, traversal methods can be used to examine all possible state transitions for verification. After excluding mutually exclusive events and conditions (e.g., $e9$ and $c10$), the considered elements

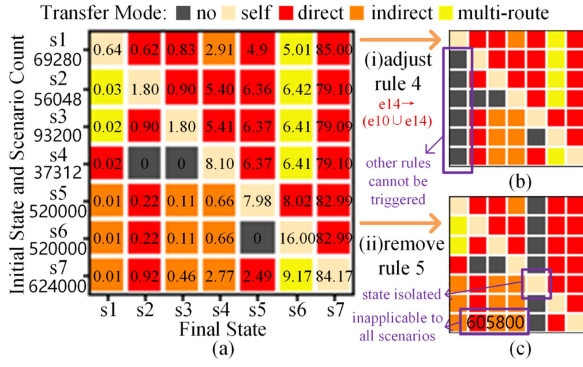


Fig. 7. Rule feasibility verification. (a) Proposed rules. (b) Rules with adjusted rule 4. (c) Rules with removed rule 5. The transition percentages are calculated based on the premise of average scenario occurrence and do not represent actual probabilities.

TABLE VIII
SIMULATION PARAMETER

Parameter	Value	Parameter	Value	Parameter	Value
L	300km	$\bar{d}_{\text{radar}}$	100km	$\bar{d}_{\text{kil}}$	0.015km
W	100km	$\bar{\theta}_{\text{radar}}$	60°	σ_p	0.2km
l	150km	$\bar{V}_{\text{radial}}$	0.04Mach	σ_v	0.04Mach
$\bar{T}$	900s	K	6	f_A	0.2hz
$\bar{h}$	15m	$\bar{d}_{\text{missile}}$	15km	M	3
$\bar{d}$	0.015km	$\bar{\theta}_{\text{missile}}$	60°	N	4
H_{upper}	20000m	H_{lower}	500m	V_{min}	0.25Mach
F_{min}	1000lb	T_{safe}	10s	$\bar{d}_{\text{fire}}$	80km

TABLE IX
CONDITIONS PARAMETER. (UNIT: KM)

	c4	c5	c6	c7	c8
s2/s4	>45	\	\	25-45(MAV) 20-45(UAV)	<25(MAV) <20(UAV)
s3	>55	50-55	45-50	25-45(MAV) 20-45(UAV)	<25(MAV) <20(UAV)

form 662,880 possible scenarios. Incorporating each state's specific requirements (e.g., c9 is incompatible with state s4), the scenario traversal confirms that the proposed rules satisfy all feasibility constraints. Fig. 7(a) presents the total number of scenarios and transition percentages for each state. As the special rules for both the MAVs and UAVs are self-state transitions, their transition situations are largely identical. Fig. 7(b) shows the infeasibility of the rules after two modifications. The total verification time is 19.79 s.

Rule effectiveness includes generalizability and performance. Generalizability requires events and conditions to capture essential elements while avoiding excessive detail. Based on distance, angle, threat, and perception, this paper's design features inherent generalization capability. Performance refers to the achievement of intended goals after actions are executed, which must be tested in confrontation. Simulation experiments in subsequent sections demonstrate that the proposed rules possess favorable performance.

Algorithm 1: Hierarchical decision-making framework.

```

input  $s(t)$ ,  $L_S$ ,  $L_E$ ,  $L_C$ ,  $L_A$ ,  $R_G$ ,  $R_{SM}$  and  $R_{SU}$ 
while Current time  $t < \bar{T}$  do
  for each of friendly fighter  $i$  do
    According to (9), target allocation is carried out to
    obtain target  $T_i$ .
    if Meet the supplementary allocation conditions then
      Adjust  $T_i$ .
    end if
    According to (11), guidance task allocation is carried
    out to obtain guidance task  $G_i$ .
    if Meet the multi-target guidance optimization
    allocation conditions then
      Adjust  $G_i$ .
    end if
    According to (13), missile launch judgment is
    carried out to obtain launch command  $Fire_i$ .
    if  $Fire_i = \text{True}$  then
      Launch a missile towards target  $T_i$ .
    end if
    Match  $s(t)$  with  $L_E$  and  $L_C$  to find the event  $E_i$ 
    currently triggered and the condition  $C_i$  that is
    satisfied.
    Obtain the status  $S_{i\_last}$  and action  $A_{i\_last}$  at the
    previous moment.
    Match  $S_{i\_last}$ ,  $E_i$ , and  $C_i$  with  $R_G$ ,  $R_{SM}$  and  $R_{SU}$ 
    to determine the rule execution plan.
    if the rule execution plan contains only one rule then
      Obtain  $A_i$  and  $S_i$  of the current rule.
    else
      Obtain  $A_i$  and  $S_i$  of the final rule.
    end if
    Execute  $A_i$ ,  $S_{i\_last} \leftarrow S_i$ ,  $A_{i\_last} \leftarrow A_i$ .
  end for
end while

```

IV. SIMULATION RESULTS

A. Simulation Setting

This paper performs a series of simulation experiments and analyses based on the scenario described in Section II. The simulation parameters are listed in Table VIII. Consistent with prior aerial confrontation research [5], [12], [16], [21], [30], this study adopts F-16 performance parameters for both MAVs and UAVs, as well as AIM-120 performance parameters for the medium-range missiles to minimize the influence of platform and weapon performance on autonomous decision-making efficacy. Based on collaborative combat aircraft costs, this paper assigns the MAVs and UAVs relative values of 1 and 0.375, respectively. The values of c4 to c8 vary depending on the type and status of the fighter. The parameters are set based on the performance of fighters and missiles, as shown in Table IX. The enemy benefits from continuous AWACS support, whereas friendly AWACS support is limited to a 200 km radius. All

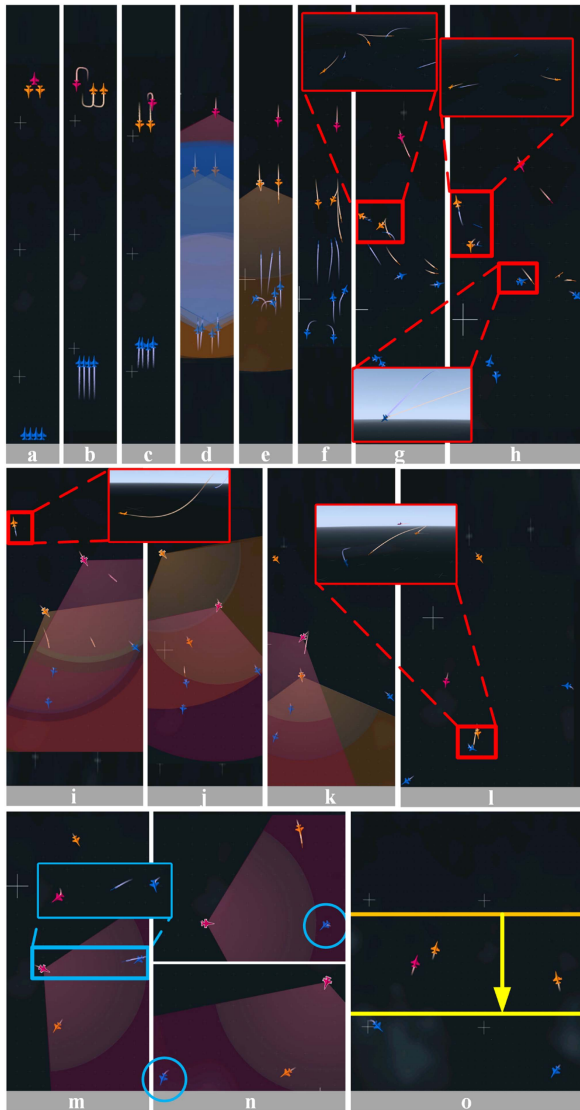


Fig. 8. Overview of the confrontation in Case 1.

simulations are implemented in Python 3.9, and the results are visualized using Tacview.

B. Typical Adversarial Case Analysis

To evaluate the applicability of the proposed framework, this study selects three representative four-fighter formation tactics as opponents, including the skate tactic (balanced), grinder tactic (defensive), and wall tactic (offensive). These tactics are adapted and improved based on the settings in [11], [12], and [21].

A detailed analysis of Case 1 is conducted, where the red side implements the proposed strategy against the blue side's skate tactic. The engagement timeline is shown in Fig. 8.

- a) Red is in cruise state (s_1), while Blue is in a horizontal formation attack state.
- b) At a large distance, Red has not detected Blue and executes a circling maneuver (a_1) according to rule 8, remaining in cruise state (s_1). Blue formation closes in rapidly.

- c) Blue enters the Red AWACS detection range. The AWACS detects and relays the information to Red fighters. Triggering "AWACS detects enemy fighter" event (e_1), Red executes a tracking maneuver (a_2) according to rule 9, transitions to attack state (s_2), and adjusts heading. Blue continues to approach.
- d) Within missile range, the FCRs of both Red and Blue detect the enemy. Blue launches missiles immediately. Red chooses not to launch due to missile strategy constraints.
- e) Blue launches multiple missiles during approach. Two fighters turn for re-entry, and others continue guidance and close in. Red UAV-1 performs a high-angle launch, followed by guidance under UAV-2.
- f) Satisfying the conditions for "beyond Phase I distance" "offensive posture" and "defensive posture" (c_1 , c_2 and c_7), UAV-1 executes a tactical missile launch (a_0) according to rule 47, followed by a 30° right offset maneuver against the fighter (a_6) remaining in attack state (s_2). Missile guidance is handed over to MAV.
- g) As distance decreases, UAV-1 and UAV-2 face missile threats. Triggering "enemy missile in terminal guidance" event (e_6) and satisfying "missile incoming angle less than 145° " condition (c_{11}), both execute a 90° right maneuver against the missile (a_3) according to rule 5, transitioning to evade state (s_5). UAV-2, unable to continue guidance, hands over the missile guidance task to MAV.
- h) Facing continuous missile attacks, UAV-2 triggers "enemy missile in terminal guidance" and "have enemy fighter" events (e_6 and $\neg e_{10}$), and satisfies "within safety assessment time" condition (c_{10}). According to rule 25, it executes a turn maneuver against the missile (a_5), transitioning to the defense state (s_6). Meanwhile, UAV-1 triggers the conditions of "no enemy missile in terminal guidance" and "safety assessment time end" ($\neg e_6$ and e_9), transitioning to attack state (s_2) according to rule 20. During this period, the missile launched by UAV-2 and guided by MAV hits an enemy fighter, while multiple Blue missiles miss.
- i) UAV-2 descends to drain missile energy, pulls up at low altitude, and breaks pursuit. After UAV-2 turns, MAV and UAV-1 split to attack in different directions and launch missiles. Although UAV-1 may face two enemy fighters, the overall situation remains controlled.
- j) After UAV-2 re-enters, MAV quickly turns to another direction to balance the number of friendly and enemy fighters on both fronts.
- k) During pursuit, MAV launches a missile and uses UAV-2's sensor for guidance, with UAV-1 in tail-chase posture. Due to degraded missile performance at low altitude and target maneuvering, the missile misses.
- l) Leveraging low-cost and expendable advantages, the UAV-1 closes in and launches a missile at closer range, hits one enemy fighter, and forces the other to turn away.
- m) After Red gains local advantage on one flank, the Blue fighter on the opposite side re-enters, posing a greater threat. With UAV-2 at a greater distance, the MAV turns

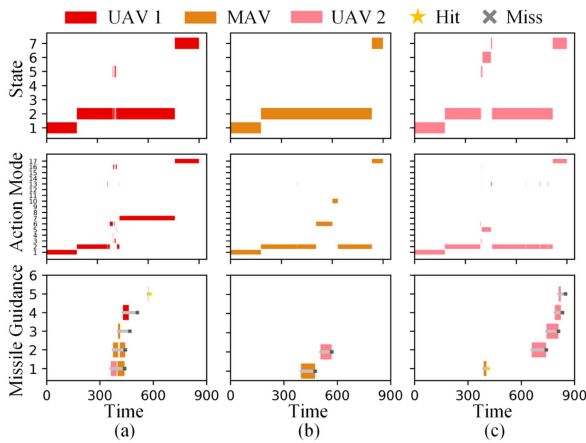


Fig. 9. Red side's tactical status, maneuvers, and missile engagement results. (a) UAV 1. (b) MAV. (c) UAV 2.

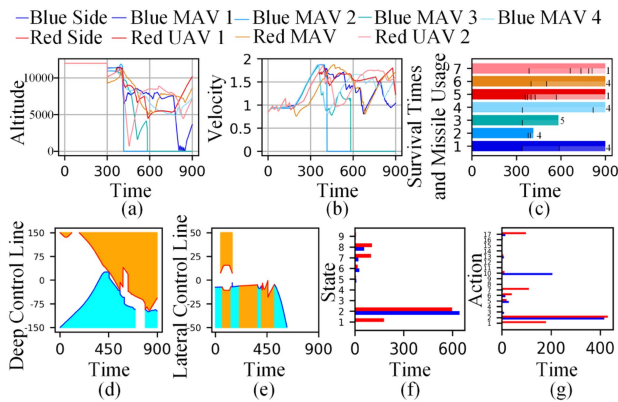


Fig. 10. Engagement situation between the two sides. (a) Altitude. (b) Velocity. (c) Survival time and missile usage. (d) Depth control line. (e) Lateral control line. (f) State time distribution. (g) Maneuver time distribution.

to engage and subsequently coordinates with UAV-2 to pincer the enemy fighter.

- n) With the situation stabilized on both sides, MAV stays at the center of the battlefield, providing continuous sensor support to both UAVs while maintaining a moderate distance from enemy fighters for safety. Although most enemy fighters fly at relatively low altitudes with continuous extension maneuvers, making them difficult to hit, Red nevertheless manages to drive them out of the battlefield and secure local air superiority.
- o) When fuel runs low, remaining fighters from both sides return to base. The depth control line changes significantly. All Red fighters retain missiles to handle other threats during return.

Remark 3: The red part of the flowchart shows the local details at the same time, the blue part shows the local comparison after some time, and the yellow arrow shows the range of changes in the depth control line. The control line is defined as the furthest position reachable by our fighters and clear of enemy presence.

Fig. 9 illustrates the tactical state transitions, maneuver selections, and missile guidance and hit conditions for each fighter on the red side. Fig. 10 illustrates the details of the engagement between the two sides.

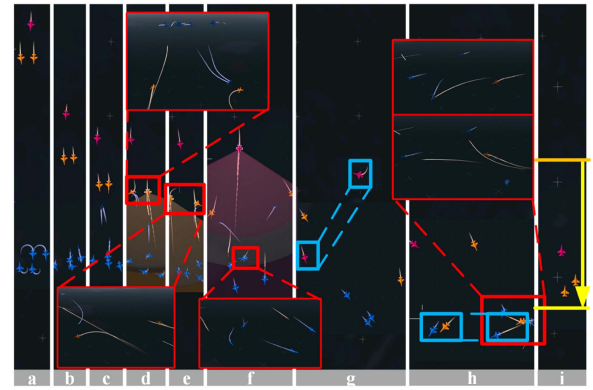


Fig. 11. Key moments in Case 2.

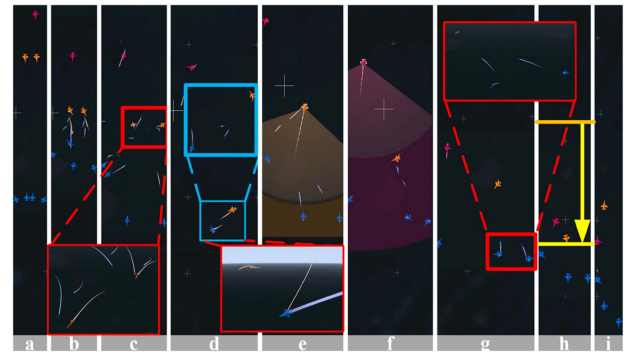


Fig. 12. Key moments in Case 3.

The data analysis indicates that the red side can rapidly respond to enemy offensives and maintain altitude and speed advantages throughout the confrontation. Regarding states and actions, the red side spends more time in attack and guidance states, and frequently employs tracking and offset maneuvers, with more weapon launch opportunities, demonstrating greater offensiveness. In contrast, the blue side spends more time in evade and defense states, and often uses turn maneuvers, adopting a passive defensive posture with longer exposure to rear threats. Regarding control lines, the red side pushes its depth control line to approximately 100 km and continuously compresses the blue side's operational space from the flanks. Throughout the confrontation, the red side sustains no fighter losses and expends 12 missiles. The Blue side loses two MAVs and expends 16 missiles (7 effectively launched, 9 consumed due to fighter destruction).

In Case 2 and Case 3, the blue side employs the grinder and wall tactics, respectively, with key engagement moments shown in Fig. 11 and Fig. 12.

Facing the highly defensive grinder tactic, the red side leverages multi-fighter coordination and threat evasion strategies to continuously deplete the blue side's missile resources, as shown in Fig. 11(d)–(f). Using UAV pincer attacks from both flanks, the red side successfully dismantles the blue side's formation, creating an encirclement and shooting down multiple enemy fighters, as shown in Fig. 11(g)–(h). Facing the highly aggressive wall formation, the red side loses one UAV due to dense missile attacks in the initial engagement, as shown in Fig. 12(b)–(c). However, through multi-fighter coordination, the red side gains

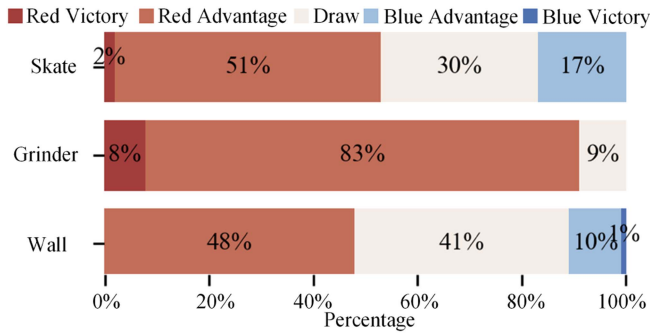


Fig. 13. Confrontation outcomes.

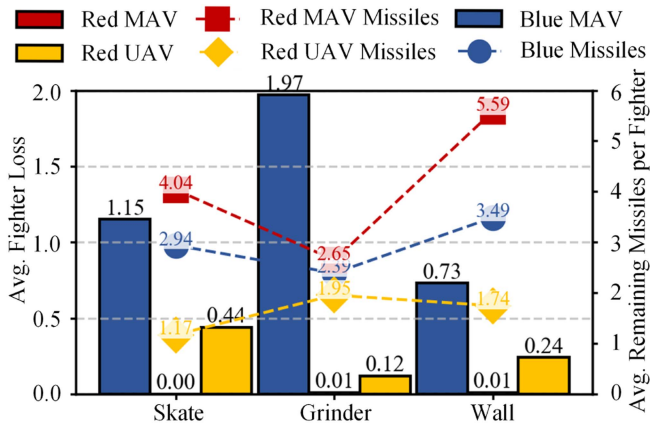


Fig. 14. Average fighter losses and average remaining missiles.

missile launch windows and completes counterattacks against the enemy, as shown in Fig. 12(d)–(e). Subsequently, the MAV and UAV fully utilize their radar resources and missile launch strategies to threaten the enemy, continuously advance the control line, and ultimately force the blue side to withdraw, as shown in Fig. 12(f)–(h).

Against blue forces employing different tactics, the red side effectively identifies enemy tactical weaknesses and dynamically adjusts combat methods to achieve superior effectiveness and expand the controllable area.

C. Statistical Analysis

To eliminate single-trial randomness, the proposed framework is evaluated in 100 repeated simulation engagements against each of the three blue-side tactics. To accelerate the testing process, parameter l is set to 100 km, with all other parameters held constant.

To comprehensively assess the effectiveness of the proposed framework, this study performs a multi-dimensional analysis. Figs. 13 to 16 present the statistical results of four aspects, corresponding respectively to the confrontation outcomes, the average losses of fighters and the average remaining missiles, the red side spatial control situation, and the frequency distribution of confrontation losses. The main data statistics are shown in Table X.

Regarding combat effectiveness, the red side consistently achieves a higher kill count against enemies employing varied

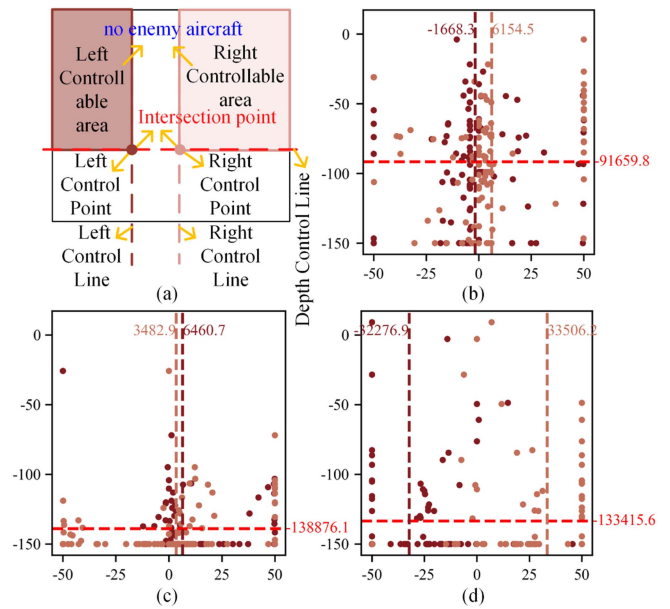


Fig. 15. Spatial control situation of the red side. (a) Explanation of control points, lines, and areas. (b) Skate opponent. (c) Grinder opponent. (d) Wall opponent.

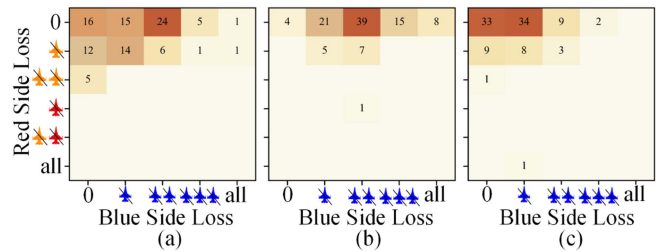


Fig. 16. Frequency distribution of confrontation losses. (a)Skate opponent. (b)Grinder opponent. (c)Wall opponent.

strategies, at times to the point of completely annihilating all enemy fighters. Regarding resource management, the red side improves missile effectiveness through coordinated guidance, eliminating more enemy fighters with fewer missiles. It also minimizes its own fighter losses via efficient defensive tactics. Regarding airspace control, the red side effectively advances the control line and compels enemy fighters to retreat, securing air superiority over a larger region.

These results validate the effectiveness of the proposed framework and its rules in complex BVR aerial confrontation. Furthermore, the enhanced survivability of the MAV increases the practical value of this framework.

D. Parameter Advantage Analysis

Inspired by [11], [12], and [21], the weight values and thresholds in this paper are fixed to ensure the interpretability of our design framework. The process to determine these final parameters is as follows: First, a user-defined baseline value for each parameter is defined, based on publicly available performance data and operational manuals for the F-16 and AIM-120. Subsequently,

TABLE X
CONFRONTATION DATA STATISTICS

Statistical Metrics	Skate	Grinder	Wall
Red win rate	2%	8%	0%
Red advantage rate	51%	83%	48%
no-damage draw rate	16%	4%	33%
Red MAV survival rate	100%	99%	99%
Red UAV survival rate	78%	94%	88%
Blue MAV survival rate	71.25%	50.75%	81.75%
Red average loss value	0.165	0.045	0.095
Blue average loss value	1.15	1.97	0.73
average value exchange ratio	1:6.97	1:43.78	1:7.68
Red MAV remaining missiles	4.04	2.65	5.59
Red UAV remaining missiles	1.17	1.95	1.74
Blue MAV remaining missiles	2.94	2.39	3.49
Red left control line	-1668.3	6460.7	-32276.9
Red right control line	6154.5	3482.9	33506.2
Red depth control line	-91,659.8	-138,876.1	-133,415.6

TABLE XI
FEASIBLE PARAMETER SETS

Type	Tag	Current	Optional 1	Optional 2
$G_t(d)$	α	(20,40,60)	(10,30,50)	(30,50,70)
$G_t(\psi)$	β	30	15	45
$G_g(d)$	γ	(40,90,100)	(30,80,90)	(50,100,110)
$G_g(\psi)$	δ	(45,60,120)	(30,45,105)	(60,75,130)
c_4-c_8	ε	(55,50,45,25,20)	(45,40,35,15,10)	(65,60,55,35,30)
G_{track}	ζ	(20,60)	(10,50)	(30,70)

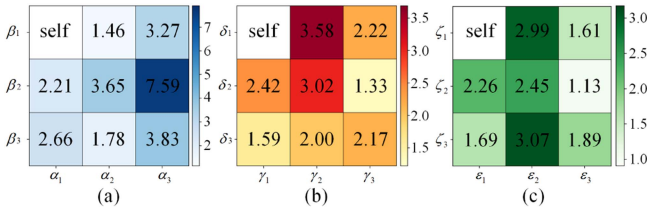


Fig. 17. Value exchange ratio of the current parameter set against each alternative set. (a) Target assignment. (b) Guidance task allocation. (c) Fire and maneuver. Data denotes the value gained per unit under the current parameters. Data from 100 runs per condition. Tag_1 , Tag_2 , and Tag_3 denote the current value and the two alternative values of the same parameter type, respectively.

these values are refined through an iterative optimization process involving extensive simulation of engagements against diverse opponent types. The iteration continues until the model achieves satisfactory operational performance, thereby yielding the final parameter values.

Remark 4: If the validation platform is changed, the parameters must be regenerated through iterative optimization based on the new platform's data.

The parameters designed in this paper primarily relate to four components: target assignment (9), guidance task allocation (11), tactical line design (Table IX), and weapon launch (13). To evaluate the superiority of the proposed parameter configuration, two additional parameter sets of distance and angle parameters are designed, as listed in Table XI. The parameters adopted in this paper are tested against feasible parameter combinations under the same group configuration (1MAV+2UAVs). The value exchange ratio is shown in Fig. 17. Results show that the proposed parameters achieve better engagement performance than the

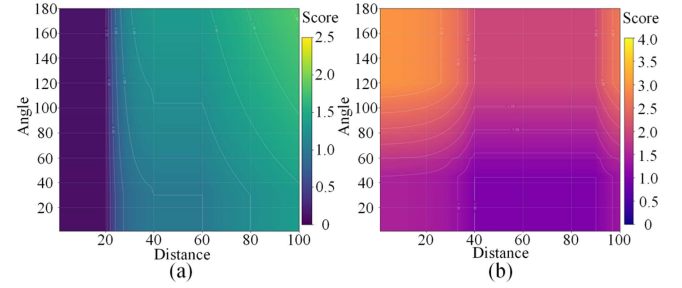


Fig. 18. Scores of current parameters across different distances and angles. (a) Target assignment. (b) Guidance task allocation.

alternative combinations. This advantage comes from the ability of the current parameters to achieve both effective utilization of weapon and sensor capabilities and a balance between offensive and defensive tactics.

The scores for target assignment and guidance task allocation obtained with the current parameters across different distances and angles are shown in Fig. 18. These results are consistent with the design expectations outlined in Section III-B.

E. Complexity and Scalability Analysis

The increase in confrontation scale often leads to higher computational complexity and timeouts, which constrains the scalability of algorithms. The proposed framework effectively manages this challenge through the hierarchical design.

In the multi-fighter collaborative layer, the computational complexity of target assignment is $O(\max(p, q)^3 + pq)$, where p is the number of surviving friendly fighters in S_2 or S_3 and q is the number of surviving enemy fighters. The complexity of guidance allocation is $O(\max(u, v)^3 + uv)$, where u is the number of surviving friendly fighters in S_2 , S_3 , or S_4 , and v is the number of missiles requiring midcourse guidance. Theoretically, the computational complexity for both target assignment and guidance allocation exhibits a cubic growth trend. In the single-fighter decision-making layer, each fighter executes predefined rule-based logic according to its assigned targets and guidance tasks. Its computational complexity, $O(c)$, depends only on the number of rules c and does not increase with the confrontation scale. The above complexity orders are expressed using Big O notation, describing the asymptotic upper bound of computational cost as key parameters grow.

Using skate-tactic opponents as the baseline, the scalability of the proposed framework is evaluated in a series of typical group configurations: 3 (1 MAV + 2 UAVs) vs. 4 MAVs, 5 (1 MAV + 4 UAVs) vs. 6 MAVs, and 7 (1 MAV + 6 UAVs) vs. 8 MAVs. Fig. 19 compares the time cost for target assignment, guidance task allocation, and fireandmaneuver decision-making across these engagement scales.

Target assignment time grows cubically with the number of fighters initially, drops after the first direct confrontation as some fighters switch to evasion, and declines further in later stages as fighter losses accumulate. Guidance allocation time rises after missile launches and fluctuates in subsequent confrontation rounds, with peak values decreasing as both fighters

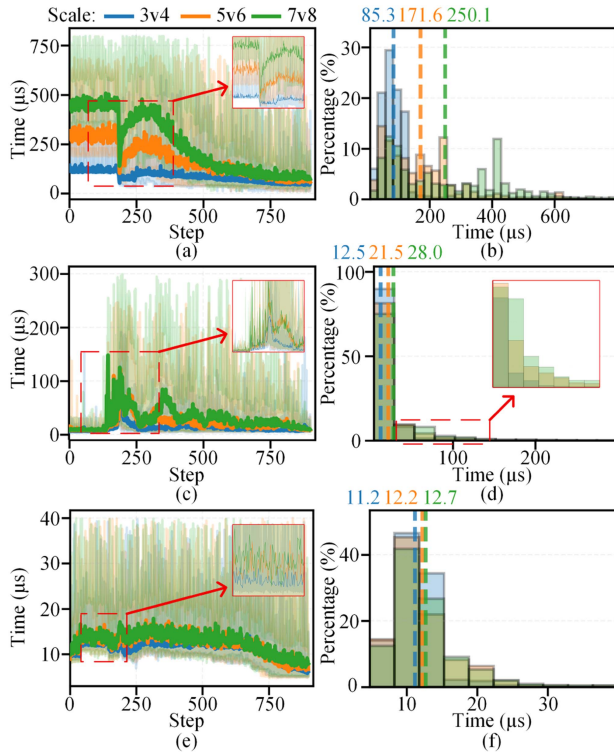


Fig. 19. Time cost across different scales. (a, c, e) Average time cost with min-max range over confrontation for target assignment, guidance allocation, and fireandmaneuver decisions, respectively. (b, d, f) Distribution and mean time cost for the same three tasks. Data from 100 runs per scale.

and missiles suffer attrition. The increase in guidance allocation time remains bounded because (14) typically assigns only two or three missiles per target. Fireandmaneuver decisions maintain nearly constant time cost across all scales and throughout the entire confrontation.

Analysis shows that the increase in the confrontation scale primarily increases the time cost in target assignment, has a moderate effect on guidance allocation, and has almost no impact on fireandmaneuver decisions. For the three single-group scales tested, the actual computation time remains within milliseconds, satisfying real-time requirements.

F. Robustness Analysis

Using enemy skate-tactic opponents as the baseline, the robustness of the proposed framework is evaluated in a series of tests under varying levels of electronic jamming and communication delay. Electronic jamming and communication delay are respectively simulated by introducing errors into the friendly side's situational awareness $s(t)$ and by delaying the update step of $s(t)$. Results in Fig. 20 compare the loss count of fighters, remaining count of missiles, value exchange ratio, depth control line, and superiority rate under varying levels of information errors and delays.

Analysis shows that our proposed framework maintains a certain superiority rate and airspace control under approximately 5% error or delays within about 5 time steps. As error increases further, threat avoidance effectiveness decreases, resulting in

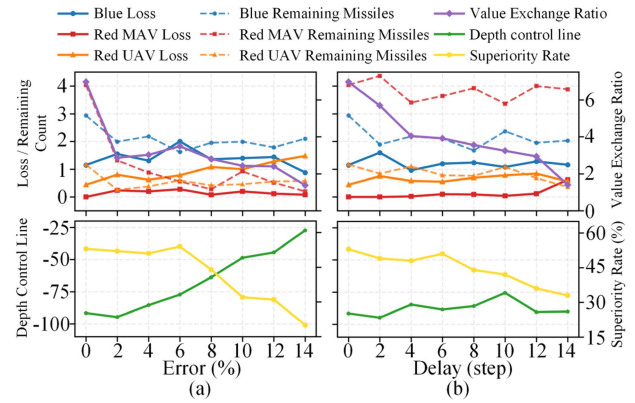


Fig. 20. Performance metrics under varying levels of errors and delays. Data from 100 runs per condition.

higher friendly fighter losses and a rapid decline in the value exchange ratio. Increased error also reduces missile hit accuracy, leading to greater missile consumption. Consequently, both the superiority rate and airspace control deteriorate sharply. Although performance degrades across all metrics as delay increases further, the decline is less pronounced and more gradual compared to that caused by information errors. This is primarily because delay has a weaker impact on threat avoidance and missile hit accuracy.

V. CONCLUSION

This paper proposes, for the first time, an SECA-driven hierarchical framework applicable to MUMT BVR aerial confrontation. The framework is fully interpretable and integrates multi-fighter collaborative decision-making with single-fighter autonomous decision-making, optimizing formation target allocation, missile guidance allocation, single-fighter maneuvering decisions, and missile launch strategies. The proposed framework fully leverages the unique advantages of MAVs and UAVs. In simulated engagements against three typical four-fighter strategies, the framework achieves strong results, with a 64% average superiority rate, a 1:16.44 average value exchange ratio, a 124,684 m average depth control line, and over 99% average MAV survival rate. Furthermore, theoretical analysis and simulation tests jointly validate the framework's validity, rationality, scalability, and robustness.

Future research will focus on perfecting the simulation platform in the sense of considering uncertainties such as electronic jamming and communication delay.

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